07 - Stacking with Random Forest and Boosting

Goal: combine text and image in a smarter way than a fixed weight.

Simple fusion (notebook 05) uses the same 70/30 weight for every product and every category. Here a second model learns how to combine them. Its input is, for each product, the 27 text probabilities and the 27 image probabilities (54 numbers).

We test two combiners:

- **Random Forest** (bagging): many decision trees trained on random samples, which then vote.
- **Gradient Boosting** (boosting): trees trained one after another, each one fixing the errors of the previous ones. We use scikit-learn's `HistGradientBoostingClassifier`, which works like LightGBM.

Honest scoring: text and image probabilities exist for the 16984 validation products, which neither model was trained on. We use 5-fold cross-validation on these products. Every product is predicted by a combiner that never saw it. The simple fusion weight is chosen the same way, so all methods are compared fairly.

1. Load text and image probabilities

```
Products: 16984
Input columns: 54
```

2. Score every method with 5-fold cross-validation

For each fold, the combiner is trained on 4/5 of the products and predicts the remaining 1/5.

Model	Weighted F1	Macro F1	Training time (s)
Image only	0.4817	0.4230	-
Text only	0.8430	0.8326	-
Simple fusion	0.8501	0.8401	-
Random Forest	0.8630	0.8491	25
Gradient Boosting	0.8595	0.8454	216

Analysis

- Both learned combiners beat the simple 70/30 fusion.
- **Random Forest is best:** weighted F1 0.8630, against 0.8501 for simple fusion (+1.3 points) and 0.8430 for text only (+2.0 points).
- Gradient Boosting is close (0.8595) but 8 times slower (about 180 seconds against 23).
- Simple fusion gives the same score as in notebook 05 (0.8501), now with the weight chosen without looking at the test products. So the old result was not biased.

3. Per category: best combiner vs text only

```
Best combiner: Random Forest
```

Code	Category	Text F1	Random Forest F1	Gain
2705	New Books	0.711	0.850	0.139
10	Used Books	0.566	0.663	0.097
2403	Book Lots	0.768	0.819	0.051
2280	Magazines	0.810	0.853	0.043
1280	Toys	0.712	0.747	0.035
40	Games	0.753	0.785	0.032
2220	Pets	0.883	0.906	0.023
1300	RC Models	0.926	0.945	0.018
2582	Garden Furniture	0.768	0.786	0.018
1160	Cards	0.954	0.971	0.017
2060	Decor	0.828	0.837	0.009
60	Consoles	0.879	0.887	0.008
1140	Figurines	0.830	0.838	0.008
2585	Garden Tools	0.833	0.839	0.006
1281	Board Games	0.616	0.622	0.006
2462	Used Games	0.825	0.829	0.005
1302	Outdoor Toys	0.837	0.841	0.004
1320	Baby	0.862	0.865	0.003
2522	Stationery	0.944	0.947	0.003
1560	Furniture	0.852	0.852	0.000
2583	Pool	0.981	0.981	0.000
2905	PC Games	0.988	0.988	0.000
1920	Textiles	0.920	0.920	-0.000
1301	Baby Socks	0.972	0.972	-0.000
50	Game Accessories	0.836	0.829	-0.007
1940	Food	0.927	0.911	-0.016
1180	Wargames	0.697	0.640	-0.057

Analysis

- The book categories gain the most: New Books +13.9 points, Used Books +9.7, Book Lots +5.1, Magazines +4.3.
- The combiner learned to trust the image more for books, which a fixed weight cannot do.
- A few categories lose: Wargames -5.7 points, Food -1.6. These are small categories (153 and 161 products), so the combiner has few examples to learn from.

Conclusion

A Random Forest that combines text and image probabilities is the best model so far: 0.8630 weighted F1 (5-fold CV on the validation products). It helps most on books, where text alone is weakest.